

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Mathematics Interactive Learning and Assessment Platform

Hackathon: Technova 2026

Domain: Software Development – Mathematics

1. Project Overview

1.1 Introduction

The **Mathematics Interactive Learning and Assessment Platform** is a web-based application developed to make mathematics learning simple, interactive, and engaging. The platform allows students to learn mathematics through structured courses, interactive activities, quizzes, graphs, and geometry tools.

Teachers can easily create courses, lessons, and quizzes without programming knowledge, while students can learn at their own pace and track their progress.

1.2 Problem Statement

Traditional mathematics learning methods mainly focus on textbooks and written exercises, making it difficult for students to understand mathematical concepts visually.

Teachers also face difficulties in creating interactive learning materials and monitoring student performance.

The proposed platform solves these problems by providing an easy-to-use learning environment with interactive activities, automatic assessments, and progress tracking.

1.3 Objectives

The objectives of the project are:

- Develop an interactive mathematics learning website.
- Allow teachers to create courses and quizzes easily.
- Help students learn through interactive activities.
- Provide automatic assessment and instant feedback.
- Track student learning progress.

- Create a responsive and user-friendly platform.
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2. Proposed System

The proposed system is a responsive web application that provides different dashboards for students, teachers, and administrators.

The platform focuses on making mathematics learning easier through interactive lessons and assessments.

3. User Modules

Student Module

Students can:

- Register and Login
- View available Mathematics courses
- Enroll in courses
- Learn through interactive lessons
- Solve quizzes and activities
- View scores and progress
- Continue learning from previous sessions

Teacher Module

Teachers can:

- Login securely
- Create new courses
- Upload lessons
- Create quizzes and assignments
- View student performance
- Monitor course progress

Administrator Module

Administrator can:

- Manage students and teachers
 - Manage courses
 - View reports
 - Maintain the system
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4. Functional Requirements

Authentication

- User Registration
- Secure Login
- Forgot Password
- Role-Based Access

Course Management

- Display available Mathematics courses
- Course enrollment
- Lesson management
- Course completion status

Interactive Learning

- Learn Algebra, Geometry, Calculus, Statistics, and Probability
- Interactive graph visualization
- Geometry drawing
- Mathematical equation editor
- Practice exercises

Assessment

- Online quizzes
- Automatic evaluation
- Instant feedback

- Hints for incorrect answers

Progress Tracking

The system records:

- Completed lessons
 - Quiz scores
 - Overall progress
 - Topic-wise performance
 - Learning history
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5. Non-Functional Requirements

- Responsive Design
 - Fast Loading
 - Secure Login
 - Easy Navigation
 - Mobile Friendly
 - User-Friendly Interface
 - Reliable Performance
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6. Website Workflow

Landing Page



Login / Register



Student Dashboard



Available Courses



Select Course



Interactive Lesson



Quiz & Assessment



Instant Feedback



Progress Dashboard

8. Expected Features

Landing Page

- Attractive homepage
- About platform
- Features
- Mathematics courses
- Login & Register

Student Dashboard

- Welcome message
- My Courses
- Continue Learning

- Progress Bar
- Quiz Scores
- Achievements

Teacher Dashboard

- Create Courses
- Upload Lessons
- Create Quizzes
- Student Reports

Admin Dashboard

- User Management
 - Course Management
 - Analytics
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9. Expected Outcome

The Mathematics Interactive Learning and Assessment Platform will provide a simple, interactive, and engaging learning environment for students.

Students can easily access mathematics courses, complete lessons, attend quizzes, receive instant feedback, and monitor their learning progress through a personalized dashboard.

Teachers can efficiently create learning materials and monitor student performance, making the teaching and learning process more effective.
